

Synthesizing Algorithms to Avoid an Obstacle with a Swarm of Robots

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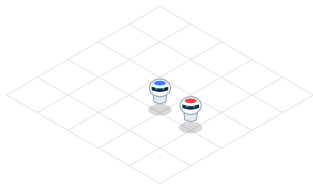
Clermont Auvergne University, France

Oussama Nahnah

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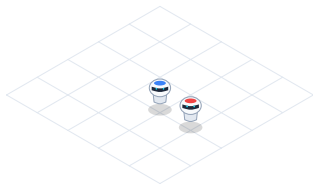
I Model: Luminous Robots

- ▶ Autonomous, Anonymous, and Homogeneous



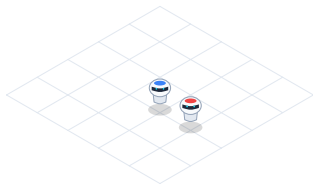
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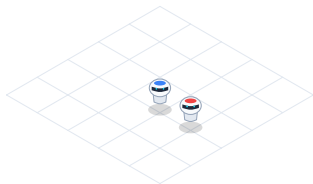
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- ▶ Operate in synchronous LCM cycles



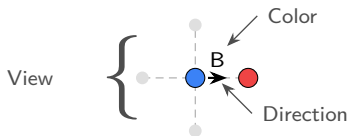
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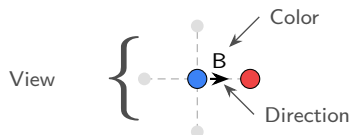
Rule components



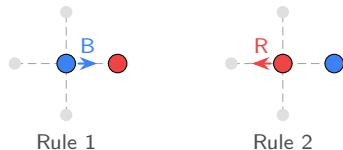
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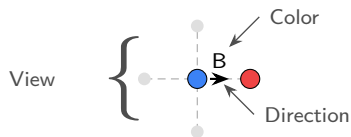
Two-rule example



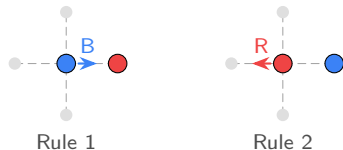
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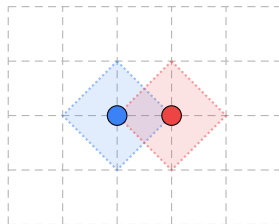
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Two-rule example



Look > Compute > Move



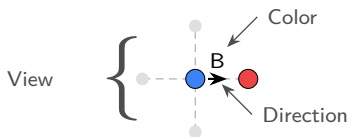
Look: Capturing local views



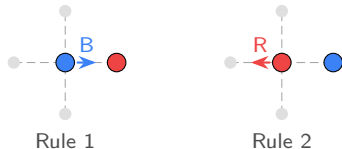
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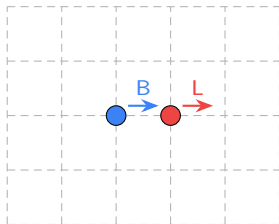
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Look > **Compute** > Move



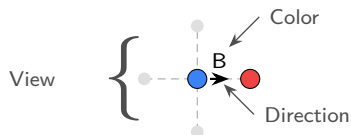
Compute: Determining next positions



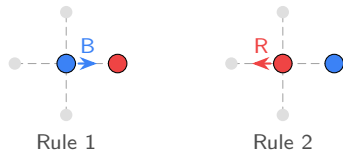
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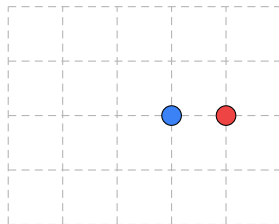
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Look > Compute > **Move**



Move: Transitioning to new position

I Model: Exploration Problem

Algo 1 [Bramas et al. 2021] 2 robots, 3 colors, visibility range $\Phi = 1$

II Related Works

► 1. The Problem: Grid Exploration

[Bramas et al. 2021, Sardar et al. 2024, ...]

→ *Limits*: Exploration without obstacle (luminous) /
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► 3. Validation: Formal Verification

[Bérard et al. 2016, Courtieu et al. 2022, ...]

→ *Limits*: Manual effort / Huge execution space

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- ▶ Perpetual exploration with obstacle (luminous, myopic, LCM cycles)

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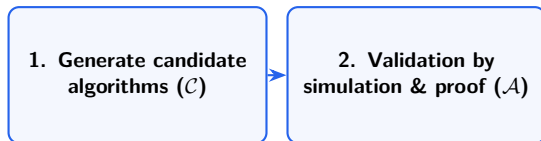
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- ▶ Partly automated method using **RoASt**

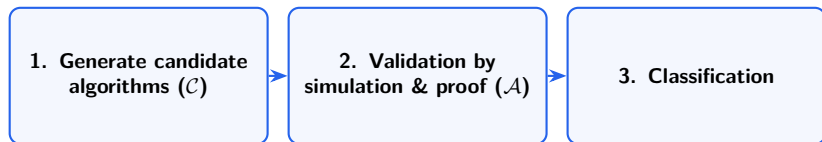
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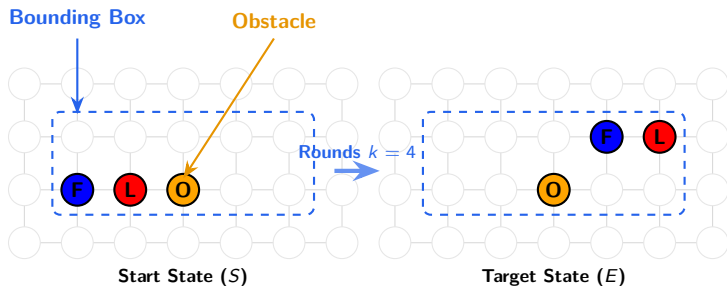


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III Phase 1: Generate candidate algorithms

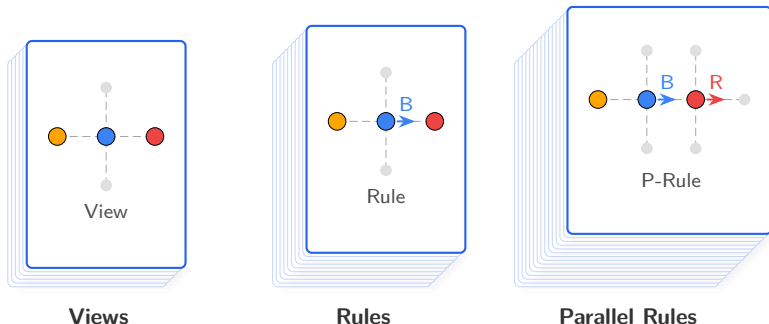
1. Define goals, bounding box, and rounds k [by hand]



III Phase 1: Generate candidate algorithms

2. Generate all possible views, rules and parallel rules

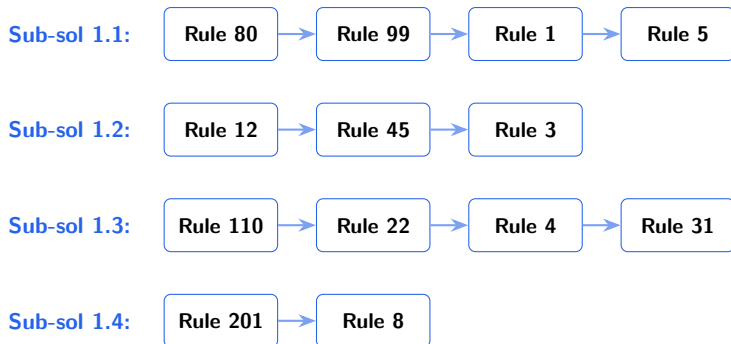
[by tool]



Pre-computed elements for Synthesis

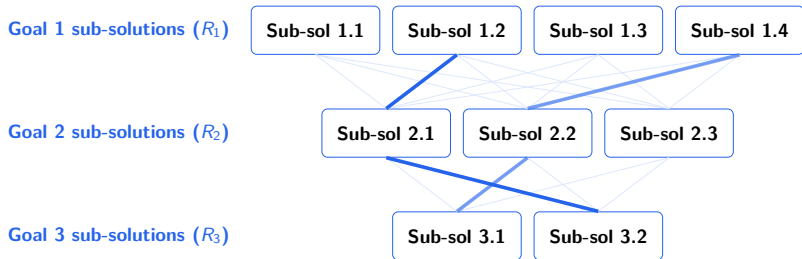
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3. Compute sub-solutions (R_i) for each goal i [by tool]



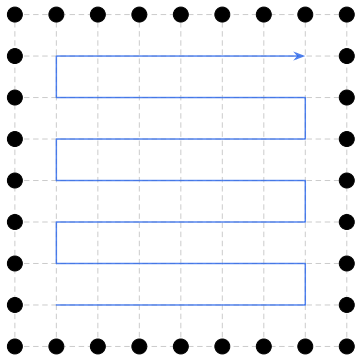
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4. Combine and filter candidate algorithms [by tool]

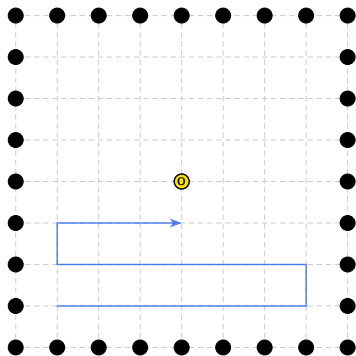


Candidate Algorithm in $\mathcal{C} = \text{Valid}(R_1 \times R_2 \times \dots \times R_m) \cup A_b$

III Phase 1: Problem Statement (Algo 1)

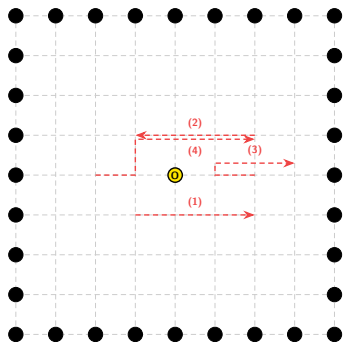


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- ▶ 4. Candidate Synthesis: 5 candidates synthesized (set $\mathcal{C}$)

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- ▶ **1. Simulation (RoASt):** candidates $\mathcal{C}$ tested on grids G_{sim}
× configs I_{sim} [by tool]

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- ▶ **2. Induction Proof (by hand):** generalize to all sizes $\implies$
validated set $\mathcal{A}$ [by hand]

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III Phase 3: Classification

Compare syntax and behavior **across all simulations** [by tool].

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- ▶ **C4: Paths in Exploration Cycle** (Simulation)
↪ Same robot trajectories in the exploration cycle on each simulation

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Results for set $\mathcal{A}$ (5 validated algorithms)

Set $\mathcal{A} = \{A_1, A_2, A_3, A_4, A_5\}$

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Conclusion: Same rounds (C2) but **5 energy-unique solutions**
(C3)

IV Other experiments

Base Algo	Scenario Variation	Candidates ($\mathcal{C}$)	Tested Levels	Synthesis Time	Valid ($\mathcal{A}$)
Algo 1	Standard Pole (8 goals)	5	4 / 4 levels	~2 seconds	5
Algo 2	Standard Pole (11 goals)	373,248	4 / 12 levels	~1 minute	115
Algo 2	See-Through Hole (Extra initial config)	373,248	4 / 12 levels	~1 minute	115
Algo 2	Goal 9 relaxed (Wider bypass distance)	117,407,808	4 / 32 levels*	3 hours 7 min	475

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- ▶ **Pole vs. Hole:** Identical outcomes
- ▶ **Combinations:** Parameter relaxation quickly yields high numbers

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- ▶ Synthesis from scratch

Thank You!

Thank you for your attention!



Scan to access the project documentation:
luminousrobots.github.io/RoASt-Docs

References

- [Bramas et al. 2021] Q. Bramas, P. Lafourcade, S. Devismes. *Optimal exclusive perpetual grid exploration by luminous myopic opaque robots with common chirality*. ICDCN, 2021.
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